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RESEARCH ARTICLE

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Oxygen Reduction

Facet-Dependent Oxygen Reduction Reaction Activity on the Surfaces of Co_3O_4

Qian Liu, Bin Yu, Xiaobin Liao, and Yan Zhao**

The mechanisms for oxygen reduction reaction (ORR) on the naturally exposed (110) and (111) surfaces of Co_3O_4 have been investigated with density functional theory (DFT) calculations. Depending on the vertical cutting place, there are type A and B surfaces for each of the (110) and (111) surfaces of Co_3O_4. Our DFT calculations reveal that the Co_3O_4 (110) type B and (111) type B surfaces have ORR catalytic activities. In addition, the ORR activity on the (110) type B surface is better than the (111) type B surface. The rate-determining step on both surfaces is thermodynamically depending on the *OH desorption process. If the solvation effects are taken into accounts, the chemically adsorbed water molecules enhance the ORR activity. According to the barrier heights calculations, $O_2 \rightarrow {}^*O_2 \rightarrow {}^*OOH \rightarrow {}^*O + H_2O \rightarrow {}^*OH \rightarrow H_2O$ route is the most favorable ORR pathway.

1. Introduction

Oxygen reduction reaction (ORR) is one of the most important steps in proton exchange membrane fuel cell (PEMFC). However, ORR is inherently complex, since it involves a four proton–electron coupled progress with adsorption and desorption of various intermediates. The sluggish kinetics originated from fully reducing O_2 into H_2O, requires large overpotential to drive reaction efficiently.[1-4] Pt-based catalysts have gain immense attention due to their outstanding electrochemical performances.[5] However, the low abundance and high cost largely restrict their further application in practice. Consequently, the development of the nonprecious catalysts such as transition-metal oxides, metal nitrides, and carbon-based materials is essential in replacing the Pt-based noble-metal catalyst.[6-9] The Co_3O_4 shows adaptive activity for ORR and has further potential applications in the Li-O_2 battery.[10-12] But it is still inferior to precious catalysts because of the limited active sites and sluggish kinetics of Co_3O_4.

The catalytic activity is significantly influenced by the atomic arrangements and electronic structures of the exposed surfaces. Co_3O_4 is spinel structure, in which 1/8 tetrahedral sites are filled by the Co^{2+} ions and 1/2 octahedral sites are occupied by the Co^{3+} ions. A thermodynamic competition between Co^{2+} and Co^{3+} exists under ambient room temperature and different partial oxygen pressure. The redox mechanism of the interconversion between different valence of Co serves as the prerequisite to the oxidation catalytic activity of Co_3O_4.[13] Moreover, the richness of active Co^{3+} sites on the Co_3O_4 (110) plane contributes significantly to the high catalytic activity for the CO oxidation.[14] Based on these observations, many researches have focused on the morphology design,[14] doping,[15] size-tuning,[16] and investigation on different exposed planes of Co_3O_4 toward water splitting and ORR, which reveal that the different electrocatalytic performances were possibly attributed to the abundance ratios of Co^{3+}/Co^{2+} over the different exposed planes.[17-19] Unfortunately, it is still uncovered that the dynamic reaction processes and the intrinsic mechanisms over different surfaces of Co_3O_4.[13,20-23]

Herein, we systematically investigate the nature of the complex surface structures and carefully identify the mechanisms of ORR on the naturally exposed (110) and (111) surfaces of Co_3O_4 by DFT calculations.[24] Two general surface termination types with the different arrangements of Co^{2+} and Co^{3+} ions for (110) and (111) planes are denoted as the type A, and type B surfaces, respectively (see SI for detailed information).[13,25] Our DFT calculations reveal that Co_3O_4 with non-metallic active sites, namely, the (110) B-1, (110) B-2, and (111) type B sites (see Section 2.2 for the definitions of these sites), shows thermodynamic preference for oxygen reduction. The overpotentials of these three active sites are 1.02, 1.02, and 1.14 V, respectively. After taking the solvation effects into consideration, the overpotentials decrease to 0.76, 0.85 and 0.83 V on the (110) B-1, (110) B-2 and (111) B sites, respectively. Our calculations indicate that the water molecule on the surface can enhance the kinetics of ORR process. Importantly, the rate-determining steps on the (110)B-1, (110)B-2, and (111)B sites thermodynamically depend on the desorption of *OH. The reaction energies and activation barriers of all possible ORR pathways have been calculated by transition state search. This work provides a microscopic understanding of the mechanisms of ORR on the Co_3O_4 surfaces and gives guidance for the design of the Co_3O_4-based ORR catalysts.

2. Results and Discussion

2.1. Catalytic Activities of the (110) and (111) Surface

In order to locate the active sites for ORR, the adsorption energies of oxygen molecules at different sites on the (110) and (111) surfaces

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have been calculated. The optimized structures and adsorption energies are shown in **Figure 1**. The (110) type A, (110) type B, and (111) type A surfaces have two kinds of active sites for the absorption of O_2, whereas the (111) type B has only one active site. The O_2 molecule adopts a bridge-site configuration on the (110) type A surface, whereas the side-site configuration is prefered on the (110) type B surface. On the (111) type A surface, O_2 adopts both bridge-site and side-site configurations, while O_2 adopting side-site configuration on the (111) type B surface. From the optimized bond lengths, it can be seen that the O-O bond lengths range from 1.330Å to 1.387 Å after adsorption, which is close to the O-O bond length of 1.33 Å in superoxide ions. However, the bond length of the O_2 molecule in the air is about 1.21 Å. These results indicate that the adsorpted O_2 molecule is significantly activated on the surface, facilitating the ORR processes.

There are two different pathways of ORR, which proceed with two-electron and four-electron transfer routes, respectively. The four-electron pathway is more desirable for ORR. Previous studies show that the four-electron pathway involves associative mechanism (Equations 1–4) and dissociative mechanism (Equations 5 and 6). The associative mechanism is employed to compare the catalytic activities of different surfaces.[26] Figures S3–S5 exhibit the structures of species adsorbed on different surfaces for each elementary reaction (Equations 1–4). The adsorption energies of all different ORR intermediates are summarized in Table S1.

$$O_2 + * + 4(H^+ + e^-) \rightarrow OOH* + 3(H^+ + e^-)0 \quad (1)$$

$$OOH* + 3(H^+ + e^-) \rightarrow O* + H_2O + 2(H^+ + e^-) \quad (2)$$

$$O* + H_2O + 2(H^+ + e^-) \rightarrow OH* + (H^+ + e^-) \quad (3)$$

$$OH* + H_2O + (H^+ + e^-) \rightarrow 2H_2O \quad (4)$$

$$1/2O_2 + * \rightarrow O* \quad (5)$$

$$O* + 2(H^+ + e^-) \rightarrow OH* + (H^+ + e^-) \quad (6)$$

$$OH* + (H^+ + e^-) \rightarrow H_2O \quad (7)$$

According to the hydrogen electrode model developed by Nørskov et al.[26], we calculate the Gibbs free energy (ΔG) for each elementary reaction at U = 0 V (**Table 1**). ΔG_1, ΔG_2, ΔG_3, and ΔG_4 represent the Gibbs free energy of reaction (1) (6) (7) and (8), respectively. From Table 1, it can be seen that ΔG_4 of (110)A-1, (110)A-2, (111)A-1, and (111)A-2 sites are positive, which indicate that reaction (4) on (110) type A and (111) type A surface is endothermic and the ORR cannot occur spontaneously at U = 0 V. These results indicate that there are no ORR catalytic activities on the (110) type A and (111) type A surfaces. As for the (110) type B and (111) type B surfaces, all elementary reactions are exothermic, and ORR can proceed spontaneously at U = 0 V. Comparing *OH adsorption energies of the type A with type B surfaces (Table S1), the *OH adsorption energies of type A are larger than those of type B. The strong adsorption energy of *OH intermediates on the type A surface may hinder the next reaction steps owing to the high energy barriers for *OH to associate with protons and electrons.

In order to determine the rate-determining step of ORR and compare the catalytic activities on the (110) type B and (111) type B surfaces, the free-energy changes of each elementary reaction at the equilibrium potential and highest potential are also calculated. **Figure 2** presents the free-energy diagrams for ORR on the Co_3O_4 surfaces. In Figure 2, the relationship U = 1.23 V (equilibrium potential) corresponds to the maximum potential of fuel cell in thermodynamics. The free energies for the formation of *O and desorption of *OH on both (110)B-1 and (110)B-2 sites go uphill, which show similar trend on the (111)B site. We can conclude that the rate-determining steps on (110)B-1, (110)B-2, and (111)B sites are the desorption of *OH, and the maximum dissociation energy is 1.02, 1.02, and 1.14 eV, respectively. The (110) type B surface shows higher catalytic activity than that of the (111) type B site. The highest potential, namely, working potential (U_{work}), is the electrode potential to keep all the elementary reactions exothermic[27,28]. The working potentials of (110)B-1, (110)B-2, and (111)B sites are 0.21, 0.21, and 0.09 V, respectively. The overpotential for ORR is defined as $\eta = 1.23\ V - U_{work}$. Base on the calculated overpotential, it is concluded that the ORR activities of these three sites are in the order of (110)B-1 ≈ (110)B-2 > (111)B. In order to verify the accuracy of the calculation, we also calculated ORR process on Pt(111) (Figure S13 and Table S4). The corresponding overpotential is 0.43V, which is basically consistent with the previous research.[26]

The project density of states (PDOSs) is shown in Figure S6, which shows that the gaps in the O_2 PDOSs disappear after the adsorption, meaning that the adsorption on the Co_3O_4 surfaces significantly changes the electronic structures of the adsorpted O_2 molecules.

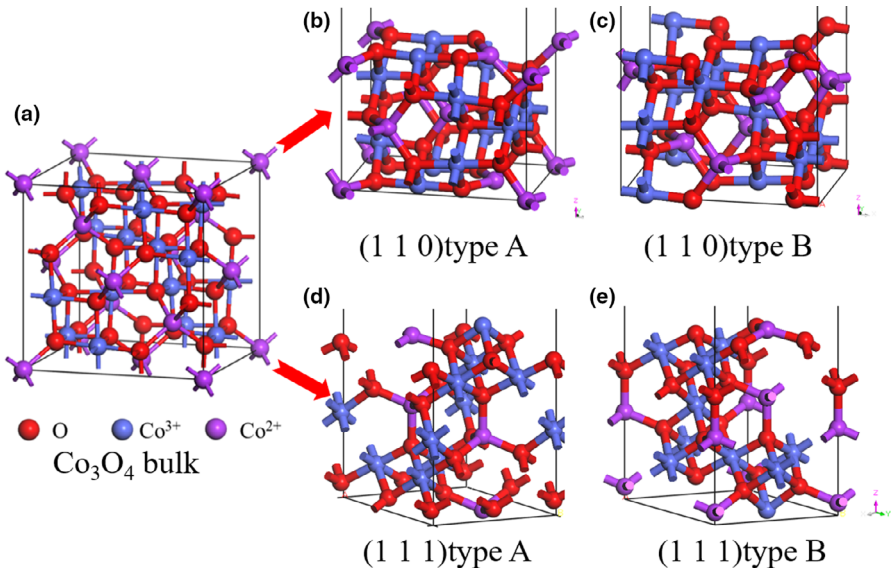


Figure 1. a) The crystal structure of bulk Co_3O_4. Co^{3+} ions is located in octahedral sites (blue) and Co^{2+} ions in tetrahedral sites (pink); b) side views of Co_3O_4 (110) surfaces in 1◇1 unit cell: type A and c) type B; and side views of Co_3O_4 (111) surfaces in 1◇1 unit cell: d) type A and e) type B.

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Table 1. The free-energy change (eV) of each elementary reaction for ORR on the surfaces at U = 0 V.

Site	ΔG_1	ΔG_2	ΔG_3	ΔG_4
(110)A-1	−2.79	−1.77	−1.22	0.86
(110)A-2	−2.79	−1.41	−0.94	0.22
(110)B-1	−2.07	−0.99	−1.65	−0.21
(110)B-2	−2.1	−0.96	−1.65	−0.21
(111)A-1	−2.54	−1.66	−1.29	0.57
(111)A-2	−2.49	−2.08	−0.96	0.61
(111)B	−2.12	−1.36	−1.35	−0.09

2.2. Solvation Effects on the ORR Activities

In order to investigate the solvation effect, water molecules are added to the (110) type B surface and (111) type B surfaces, as shown in Figure S7. Figures S8 and S9 present the top and side views of intermediates for the ORR reactions. As shown in Figure S9, the solvent water molecules are chemically adsorbed on the active site due to the strong interactions with the Co atoms. This is consistent with a recent mechanic study of ORR on Pt(111) by Liu *et al.*[29] They have found that the chemically absorbed water molecule is important for the reconciliation of the discrepancy of theory and experiment.

Figure S8 shows that the O-O bond length ranges from 1.281 to 1.344 Å, and the adsorption energies of the intermediates are listed in Table S2. If we compare the results with and without water molecules (Table S1 vs Table S2), we can see that the average adsorption energies are smaller with the presence of water molecules, meaning that the water molecules weaken the chemisorption of ORR intermediates on the surfaces. According to the Sabatier's principle, the interactions between the sorbate and catalyst should be neither too strong nor too weak.[26,30] Our calculations demonstrate that the water solvation effect improves the catalytic activity on the Co_3O_4 (110) type B and (111) type B surfaces.

We also calculate the Gibbs free energy (ΔG) for each elementary reaction in the presence of water molecules. The free-energy change of each elementary reaction for ORR at U = 0 V are listed in Table S3, and the free-energy diagrams are shown in Figure S10. The free energies of formation of *O and desorption of *OH on the (110)B-1,

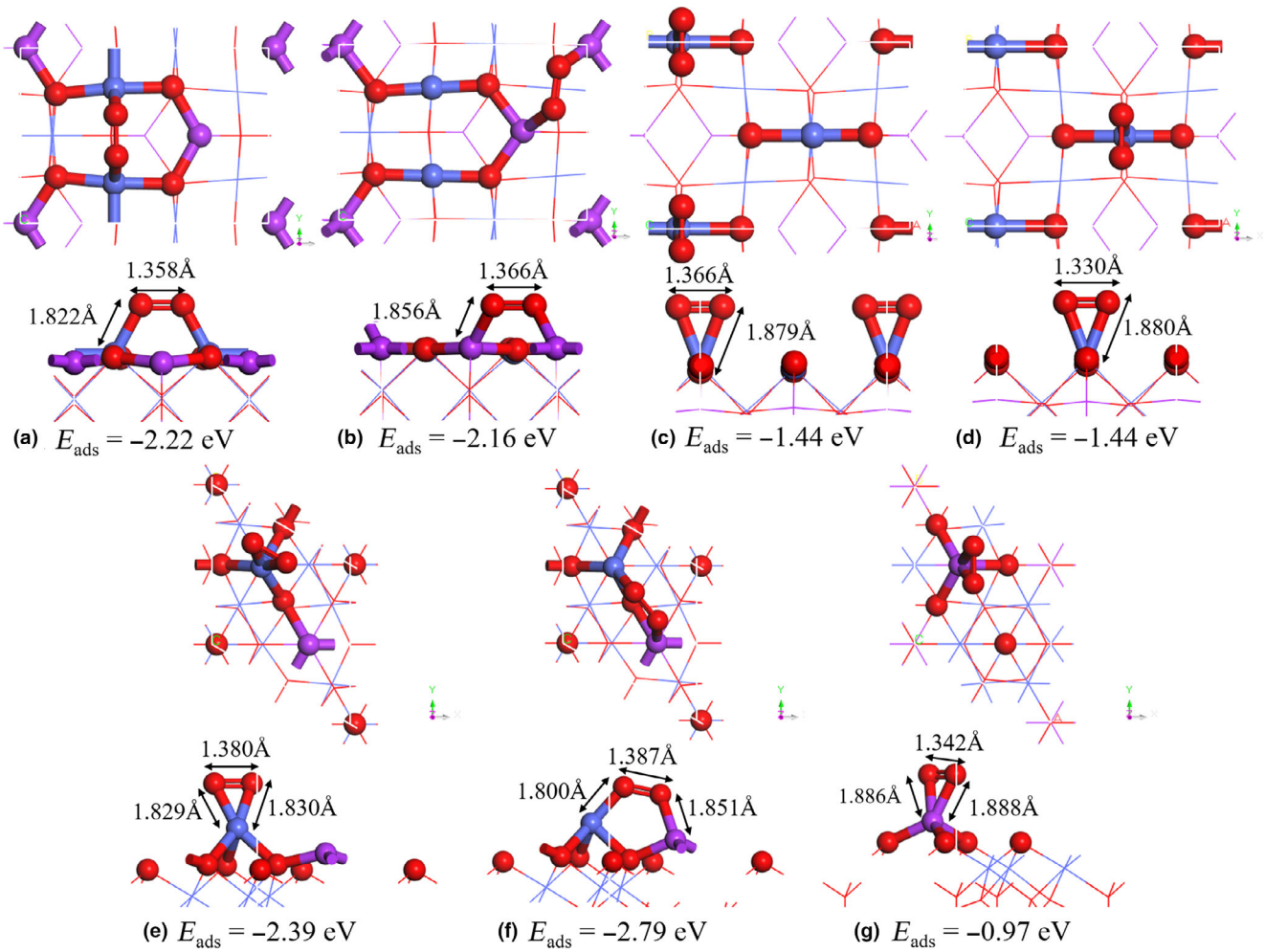


Figure 2. Top and side views of O_2 adsorption structures on (110) surfaces. a) (110)A-1 site; b) (110)A-2 site; c) (110)B-1 site; and d) (110)B-2 site. Top and side views of O_2 adsorption structures on (111) surfaces. e) (111) A-1 site; f) (111)A-2 site; and g) (111)B site.

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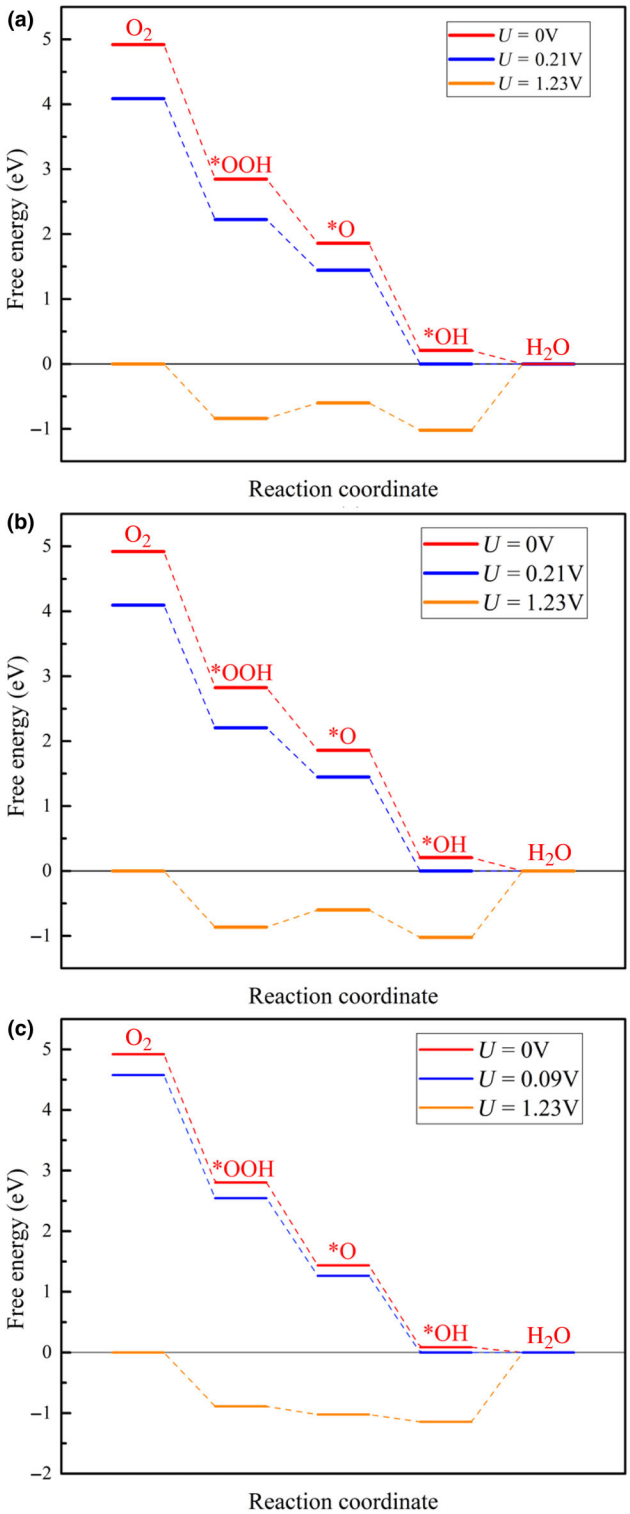


Figure 3. Free-energy diagrams for the ORR on Co_3O_4 surfaces. Results are shown at zero cell potential (red line), at the equilibrium potential (yellow line), and at the highest potential (blue line). a) (110)B-1; b) (110)B-2; and c) (111)B.

(110)B-2, and (111)B sites go uphill. However, without the water molecules, the step of *OH desorption on (111)B goes uphill. Comparing Figures 2 and S10, the rate-determining steps on (110)B-1, (110) B-2, and (111)B sites are the *OH desorption step. Nevertheless, the ΔG of the rate-determining steps is significantly decreased (Figure S10), which are 0.76, 0.85, and 0.83 eV for (110)B-1, (110)B-2, and (111)B sites, respectively. The corresponded highest potentials with solvation effects for (110)B-1, (110)B-2, and (111)B are 0.47, 0.38, and 0.40 V, respectively. These results reveal that the surface water molecules can improve the ORR catalytic activity of the Co_3O_4 by weakening the interaction between the intermediates and active sites, and facilitating each step of ORR reactions.

2.3. Transition State Search on the Co_3O_4 (110) type B Surface

From the results of previous sections, the (110)B-1 site on the Co_3O_4 (110) type B surface has the best ORR catalytic activity. Thus, in this section, we employed (110)B-1 site to find the most kinetically favorable ORR process. The possible reaction pathways with reaction barrier and reaction energy are summarized in **Figure 3**. The details of O_2 hydrogenation (a_2), formation of *O (c_5), hydrogenation of *O (d_1), and desorption of *OH(e_1) are shown in Figures S11 and S12.

As shown in Figure 3, the most favorable process for the O_2 dissociation pathway is $a_1 \rightarrow b_1 \rightarrow c_1 \rightarrow d_1 \rightarrow e_1$, the rate-determining step (highest activation barrier) of which is O_2 dissociation step (a_1). As for the O_2 hydrogenation pathway, the $a_2 \rightarrow c_5 \rightarrow d_1 \rightarrow e_1$ pathway is the most favorable and the rate-determining step is a_2 (Figure 3). Comparing the activation barrier of rate-determining step for O_2 dissociation (1.62 eV) and hydrogenation pathway (0.44 eV), the O_2 hydrogenation pathway ($a_2 \rightarrow c_5 \rightarrow d_1 \rightarrow e_1$) is more favorable than the O_2 dissociation pathway. A previous study[31] has shown that the most favorable ORR process on Pt (111) (*O + *OH + *H→ *OH + *OH) has the activation barrier of 0.86 eV. The Co_3O_4 (110) type B surface has a much lower kinetic activation barrier as compared to Pt (111). However, the overpotentials of pure Co_3O_4 (110)B-1, (110)B-2, and (111)B sites are 1.02 V, 1.02 V, and 1.14 V, higher than Pt (111) (0.43 V), meaning Co_3O_4 has an inferior catalytic activity for ORR as compared to Pt from the thermodynamic perspective. Nevertheless, the low-cost, high-stability, and low kinetic activation barriers make Co_3O_4 a promising candidate for a good ORR catalyst.

3. Conclusions

In this work, DFT was employed to investigate the electrocatalytic ORR performances of Co_3O_4 (110) & (111) planes. Our DFT calculations reveal that the Co_3O_4 (110) type A and Co_3O_4 (111) type A surfaces do not have ORR catalytic activity. As for the Co_3O_4 (110) type B and (111) type B surfaces, the overpotentials of the pure Co_3O_4 (110)B-1, (110)B-2, and (111)B sites are 1.02, 1.02, and 1.14 V, whereas when considering solvation effects, the overpotentials decrease to 0.76, 0.85, and 0.83 V, respectively. These theoretical analyses reveal that water on the surface can enhance the kinetics of ORR process. The rate-determining step on the (110)B-1, (110)B-2, and (111)B sites is the desorption of *OH, whether considering the solvation effects or not. According to the DFT calculations, the presence of H_2O will weaken the adsorption ability of intermediates on both the (110) type B and (111) type B surface. The possible reaction pathways on the promising Co_3O_4 (110)B-1 site were investigated systematically. The O_2 hydrogenation pathway is more kinetically favorable than O_2 dissociation pathway. The $O_2 \rightarrow *O_2 \rightarrow *OOH \rightarrow *O + H_2O \rightarrow *OH \rightarrow H_2O$ pathway is

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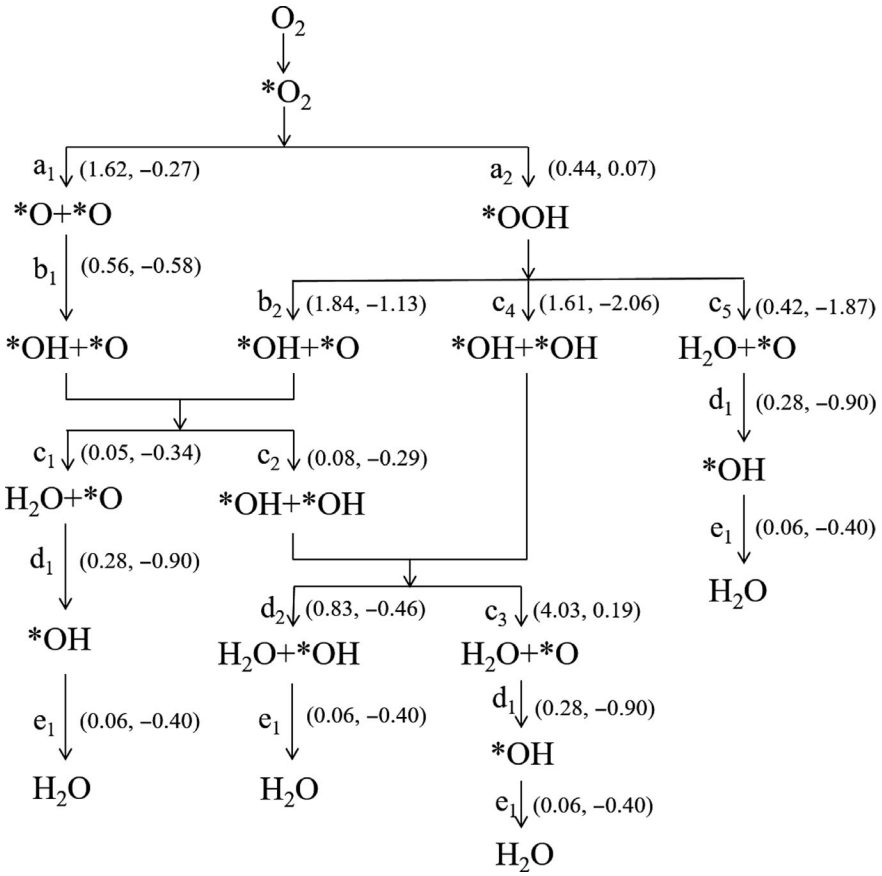


Figure 4. a) The crystal structure of bulk Co_3O_4. Co^{3+} ions is located in octahedral sites (blue) and Co^{2+} ions in tetrahedral sites (pink); b) side views of Co_3O_4 (110) surfaces in 1◇1 unit cell: type A and c) type B; and side views of Co_3O_4 (111) surfaces in 1◇1 unit cell: d) type A and e) type B.

the most favorable ORR pathway due to lower activation barriers, and the rate-determining step is $*O_2 \rightarrow *OOH$. We expect our study can provide a basis for further theoretical and experimental studies of Co_3O_4 based catalysts for ORR.

4. Experimental Section

Computational Methods: The DMol^3 program[32,33] was employed to perform the DFT calculations with the Perdew−Burke−Ernzerhof (PBE) generalized gradient approximation (GGA) functional.[34] The double numerical plus polarization (DNP) basis set was used to describe the atomic orbitals,[35] and the DSPP semi-core pseudopotential was adopted to describe the core electrons of the transition-metal atoms, in which the core electrons of Co were replaced by a single effective potential and the outer electrons ($3d^74s^2$) were treated as valence electrons.[36] A real-space global orbital cutoff radius of 4.5 Å was employed to ensure the convergence. A thermal smearing of 0.01 hartree was chosen to improve the self-consistent field (SCF) convergence. The convergence criteria are the following: 1) an SCF tolerance of 10^{-6} au; 2) an energy change of 10^{-5} hartree; 3) a max force tolerance of 0.002 hartree/Å; and 4) a max displacement tolerance of 0.005 Å. The empirical dispersion correction of the Grimme scheme[37] was applied to describe the Van der Waals interactions.[38] The frequency calculation has been done for the bulk Co_3O_4 primitive cell. The spin-unrestricted scheme was utilized throughout the calculation. The Brillouin zone integrations were sampled using a Monkhorst-Pack 5 × 5 × 5 grids for bulk, 5 × 5 × 1 for the (111) surface and 3 × 4 × 1 for the (110) surface. The ORR minimum energy paths (MEPs) and transition state search were obtained by the complete LST/QST method.[39,40] The eigenvector-following method was adopted to optimize the structures of all species and transition states.[41]

The adsorption energy (E_{ads}) for the species in the ORR process is calculated as follows:

$$E_{ads} = E_{adsorbate+slab} - E_{slab} - E_{adsorbate} \quad (8)$$

where $E_{adsorbate+slab}$, E_{slab} and $E_{adsorbate}$ are the total energies of slab model with adsorbates, clean slab, and free adsorbate, respectively.

When solvation effects are taken into consideration, the adsorption energy is calculated as:

$$E_{ads} = E_{adsorbate+slab+H_2O} - E_{slab+H_2O} - E_{adsorbate} \quad (8.1)$$

The activation barrier E_a is defined as the energy difference between the reactants and transition state (TS):

$$E_a = E_{TS} - E_{reactants} \quad (9)$$

The energy of reaction is determined as:

$$\Delta E = E_{products} - E_{reactants} \quad (10)$$

where $E_{reactants}$, E_{TS}, and $E_{products}$ are the total energies of reactants, transition states, and products, respectively.

The free-energy diagrams were calculated by the approach developed by Nørskov et al.,[26] and ΔG is evaluated as:

$$\Delta G = \Delta E + \Delta ZPE - T\Delta S + \Delta G_U + \Delta G_{pH} + \Delta G_{field} \quad (11)$$

where ΔE, ΔZPE, and ΔS are the energy of reaction, the zero-point energy contribution, and entropy change, respectively. T is the temperature (set to 298.15 K). $\Delta G_U = -eU$ is the free energy of a single proton − electron pair relative to H_2 in the gas phase at standard conditions, where U is the electrode potential with respect to the reversible hydrogen electrode. ΔG_{pH} is the correction of the H^+ free energy, which can be calculated through $\triangle G_{pH} = k_B T \times \ln 10 \times pH$. The k_B is the Boltzmann constant, and pH is set to 0 (acidic medium). ΔG_{field} is the free-energy correction resulted from the electrochemical double layer, which is negligible in this study.

Bulk and Surface Models of Co_3O_4: The bulk Co_3O_4 (a = b = c; $\alpha = \beta = \gamma = 90°$) was optimized through two steps. Firstly, a quadratic polynomial function was fitted for the total energy and lattice parameter. Figure S1 shows the quadratic function curve fitting for the total energy vs lattice constant, and the optimum lattice parameter is 8.063 Å. The atomic coordinates were further optimized with the lattice parameter of 8.063 Å. All the calculations were performed with the initial spin values of 3 for Co^{2+} and 0 for Co^{3+},[25] and the calculated frequencies (Supporting Information) for Co_3O_4 are positive, demonstrating the good stability of the Co_3O_4 catalyst.

The (110) and (111) planes of Co_3O_4 have two types of terminations.[13] On the type A surface (**Figure 4**b) of Co_3O_4 (110), Co^{2+} and Co^{3+} are both exposed with 3-fold coordinated oxygen atoms. There are only Co^{3+} cations with 2-fold and 3-fold coordinated oxygen atoms on the Co_3O_4 (110) type B surface (Figure 4c).[25] During the relaxations of the two surfaces, top two layers were relaxed and the other 3 layers were fixed. The Co_3O_4 (111) surface is quite different from (110) surface, and the constraint condition of (111) surfaces is shown in Figure S2

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of Supporting Information. The type A surface of (111) (Figure 4d) is terminated with Co^{3+}, whereas the type B surface (Figure 4e) is terminated with Co^{2+}. According to the previous report, we increased the number of layers to maintain the antiferromagnetic properties of Co_3O_4.[25]

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Conflicts of Interest

There are no conflicts to declare.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Keywords

Co_3O_4, density functional theory, oxygen reduction reaction

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